Homework 5 (65 pts)

Instructions

For this homework you will create a gitHub repo and save a python notebook (`.ipynb` file) using Google Colab there. You'll then submit a link to your gitHub repo. Be sure to:

- change the sharing settings on your notebook. This can be done by clicking on the 'share' icon in the top right. Please make sure to change the sharing settings so that anyone with the link can see the notebook.
- **include markdown text describing what you are doing, even when not explicitly asked for!**
- **save updates you make to your notebook to gitHub as you work through it to earn full credit.** No edits should be made after the due date!

The purpose of this homework is to explore issues around big data including bias in data and models along with practicing SQL skills.

Part I - Concepts (11 pts)

1. We discussed the "Five V's of Big Data". Give an example of a place where you've encountered big data or a topic you are interested in where big data would arise. Specifically address the five V's for the example and whether/how they apply to your example. (5 pts)

2. Read about the database we'll use below (part IV). Give an example for each letter in the `CRUD` acronym in the context of this database. (3 pts)

3. What is the purpose of the `HAVING` clause when writing SQL code? (Recall the sqlitetutorial.net tutorial on the `SELECT` statement.) (3 pts)

Part II - Simulation of a Sampling Distribution (20 pts)

4. We looked at using simulation to investigate the sampling distribution of $\hat{p}$ in the notes. We'll now look at the sampling distribution of the sample slope from an SLR model using simulation.

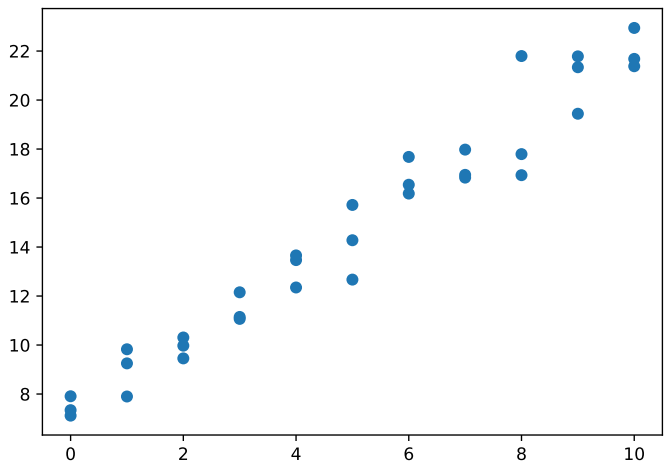
   - Recall we assume the following model for SLR:

     $$Y_i = \beta_0 + \beta_1 x_i + E_i$$

     where the E_i are assumed to be indendent and identically distributed from a Normal distribution with mean 0 and variance σ^2. Let's assume $\sigma^2 = 1$ for simplicity.

   - We can generate data from this model by assuming values for β_0, β_1, and n, along with a sequence of x values via the following code:

```python
#import some modules needed
import matplotlib.pyplot as plt
import numpy as np
from numpy.random import default_rng
rng = default_rng(32)
beta_0 = 7
beta_1 = 1.5
# get three 'values' of x at each integer from 0 to 10.
x = np.array(list(np.linspace(start = 0, stop = 10, num = 11))*3)
n = 33
#create the 'responses' modeled from the line plus a random deviation
y = beta_0 + beta_1*x + rng.standard_normal(n)
#visualize the data
plt.scatter(x = x, y = y)
plt.show()
```



- Now we can use `sklearn` to obtain the estimate for the slope (and save that value) as we did earlier in the course.
- Repeat the above process 5000 times. To be specific: (11 pts)
  - Generate the y values using the code with `rng.standard_normal()`
  - Find the 'best' slope via `sklearn`
  - Save that slope in some object
- We can use the many values of the sample slope as an approximation to the sampling distribution of the sample slope! Create a histogram of the sample slope values you found. (3 pts)
- Use your sampling distribution to approximate the probability of observing a sample slope larger than 1.65. This can be done by looking at the proportion of simulated values larger than 1.65. (3 pts)
- Give an interpretation of this value and why it might be important for us in relation to a hypothesis test. (3 pts)

Part III - Big Data Examples & Rare Events (14 pts)

5. Consider this article about bias in machine learning models

- Pick an example in the article and briefly outline the issue around the data used to fit the model. (3 pts)
- Describe how we might augment the data or otherwise account for the bias in that situation. (3 pts)

6. Consider this article about big data usage in the Earth Sciences
   - Choose one of the three major areas the paper discusses. Summarize the use of big data there. Are there any biases or other missing data that might affect results? (I know this is likely difficult as most of us aren't experts in Earth science - just do your best - my goal is to try to get us to always ask questions about the data collected and used for a study!) (5 pts)
7. Read through this short article: Article
   - Mostly I just want you to read this article to understand some ways in which rare events, and statisticians, play a role in different places. To have something to put here, can you think of another variable or scenario where understanding rare events is important? (3 pts)

Part IV - Querying a database (20 pts)

There is a database file on the assignment link called Lahman.db that is an sqlite database downloaded from here. This database has information on Major League Baseball.

8. Connect to the database (probably by uploading to Colab) and then look at all of the tables in the database (use `read_sql()` from `pandas` to have this returned as a data frame). (2 pts)
9. Write an SQL query using `pd.read_sql()` that returns all the teams that played in the year 2015 with all of the corresponding columns from the `Teams` table. (2 pts)
10. Write an SQL query using `pd.read_sql()` that returns all of the players in the hall of fame, the year they were voted into the hall of fame, and their category - see the `HallOfFame` table, the inducted variable is important here. (3 pts)
11. Write an SQL query using `pd.read_sql()` that return all unique managers of the Pittsburgh Pirates (`teamID` of `PIT`) and only that information from the `Managers` table. Hint: Check out `SELECT DISTINCT` (3 pts)
12. Use SQL code and the `HallOfFame` and `Managers` tables to return all of the `playerIDs` for the people that managed for a team that were inducted into the hall of fame. Also, programmatically report the number of such people - this can be done in `pandas` after returning the data from the call to `pd.read_sql()`. (4 pts)
13. Now use the same two tables (`HallOfFame` and `Managers`) and an SQL query to return every season managed by each manager that made it to the hall of fame (so do this with an SQL join, not via `pd.merge()`). You should return the `playerID` (manager ID), `G`, `W`, and `L` columns from the `Managers` table.
    Second, determine the overall win/loss records (sum of wins and sum of losses) for each of these hall of fame managers. Third, create a new variable that is the win/loss percentage (`W/(W+L)`). Lastly, sort the resulting data by the win/loss percentage variable (from largest to smallest). The last three parts can be done in `pandas` with the returned data or you can do it via SQL in your call to `pd.read_sql()`. (6 pts)

Submission

Now you can grab the link to your gitHub repo and submit that! Good luck and let us know if you run into issues.